Let $P(x)$, $Q(x)$ and $R(x)$ be the statements "x is a lion", "x is fierce", and "x drinks coffee," respectively. Assume that the domain consists of all creatures, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$ and $R(x)$.

Soln: Statement 1: $\forall x(P(x) \rightarrow Q(x))$

Statement 2: $\exists x(P(x) \land \neg R(x))$

Statement 3: $\exists x(Q(x) \land \neg R(x))$

Notice that the second statement cannot be written as $\exists x(P(x) \rightarrow \neg R(x))$. The reason is that $P(x) \rightarrow \neg R(x)$ is true whenever x is not a lion, so that $\exists x(P(x) \rightarrow \neg R(x))$ is true as long as there is atleast one creature that is not a lion, even if every lion drinks coffee. Similarly, the third statement cannot be written as $\exists x(Q(x) \rightarrow \neg R(x))$.

Example 27: Consider these statements of which the first three are premises and the fourth is a valid conclusion.

"All hummingbirds are richly colored."

"No large birds live on honey."

"Birds that do not live on honey are dull in color."

"Hummingbirds are small."

Let $P(x)$, $Q(x)$, $R(x)$ and $S(x)$ be the statements "x is a hummingbird." "x is large," "x lives on honey," and "x is richly colored," respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$ and $S(x)$.

Soln: Statement 1: $\forall x(P(x) \rightarrow S(x))$

Statement 2: $\forall x(Q(x) \rightarrow \neg R(x))$

Statement 3: $\forall x(\neg R(x) \rightarrow \neg S(x))$

Statement 4: $\forall x(P(x) \rightarrow \neg Q(x))$